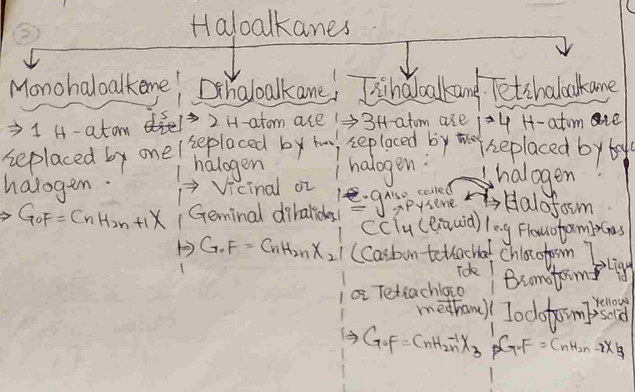


Alkyl Halide

Halogen derivative of alkane are called haloalkane. OR

Compounds in which one or more than one hydrogen of alkane are replaced by halogen are called haloalkane.



Haloalkanes are also called Organohalogen compounds.

Monohaloalkanes are generally called alkyl halides.

The F.G of alkyl halides is halogen with G.R of $R-X$.

The G. formula of alkyl halide is $C_n H_{2n+1} X$.

$R-X$ is most suitable to incorporate R-group in organic compounds.

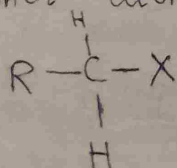
Alkyl halides are;

- ★ Artificial compounds.
- ★ Oily in nature
- ★ Saturated compounds.

Types of Alkyl Halide

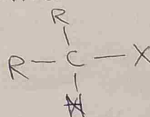
(1) 1° Alkyl halide:

The type of alkyl halide in which halogen bearing carbon is attached to one further atom:



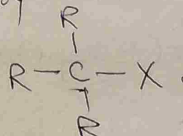
(2) 2° Alkyl Halide:

Alkyl halide in which halogen is attached to 2ndary carbon atom.



(3) 3° Alkyl Halide:

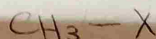
Alkyl halide in which halogen is attached to 3ary carbon atom.



$\Rightarrow$ It has no H-atom with the carbon atom.

(4) Methyl halide (Halomethane):

When halogen is attached to methyl gp.



MCQ

W.O.F alkyl halide is not possible?

- (a) 1° (b) 2° (c) 3° (d) 4°

(10) All $R-X$ (RF, RCl, RBr) can be prepared directly except RI .

(11) It is bcz of the reason that RI is strong reducing agent.

(12) When halogen atom is bonded to sp^3 hybridized carbon of alkane are called Alkyl halides.

(13) When halogen atom is bonded to sp^2 hybridized orbital are called Vinyl halide.

(14) When halogen atom is attached to aromatic ring (benzene) is called Aryl halide.

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When halogen atom is attached to sp -hybridized carbon atom is called acetylene halide.

* CH_3CH_2-Cl (Alkyl halide)

* $CH_2=CH-CH_3$ (Vinyl halide)

* $CH \equiv C - \begin{matrix} Cl \\ | \\ Cl \end{matrix} - CH_3$ (Acetylene halide)

*  (Aryl halide)
or Haloarene

*  (Alkylcyclic halide)

*  (Benzylic halide)

Structure of methyl chloride

In methyl chloride, carbon atom is sp^3 hybridized.

The geometry of methyl chloride is tetrahedral.

There are two types of sigma bonds in methyl chloride.

(a) sp^3 -p orbital overlapping.

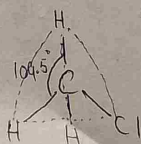
(b) sp^3 -s orbital overlapping.

There are four sigma bonds & no pi bond in methyl chloride.

C-H-C & C-H-Cl bond angle is 109.5° .

The C-Cl bond length is 178 pm or 1.78 Å.

The C-H bond length is 108 pm.



methyl chloride

Physical properties of alkyl halides

Phenyl halides

methyl fluoride, CH_3Cl , CH_3Br , C_2H_5F , C_2H_5Cl , C_3H_7F are gases

$\Rightarrow C_4 - C_{16}$ are liquids

$\Rightarrow C_{17}$ & so on are solids.

B.P.

$\Rightarrow$ They have higher boiling pt than the corresponding alkanes due to polarity.

$\Rightarrow$ For a given alkyl group, the B.P. increases with increase in size of halogen due to $\uparrow$ VWF.

Size of halogen $\propto$ B.P.

$\Rightarrow$ For a given alkyl group, the B.P. increases with increase in size of alkyl group due to $\uparrow$ in VWF.

Size of alkyl group $\propto$ B.P.

$\Rightarrow$ With increase in number of halogen atom, B.P. increases.

(a) $CH_3-I > CH_3-Br > CH_3-Cl > CH_3-F$

(b) $C_4H_9-F > C_3H_7-F > C_2H_5-F > CH_3-F$

(c) $CCl_4 > CHCl_3 > CH_2Cl_2 > CH_3Cl$

$\Rightarrow$ For isomeric alkyl halide, the B.P. of the alkyl halide with more branches will be lower.

No of Branching $\propto \frac{1}{B.P.}$

(a) $CH_3CH_2CH_2CH_2Cl > CH_3-\overset{\overset{CH_3}{|}}{CH}-Cl$

(b) $CH_3CH_2CH_2CH_2Cl > CH_3-\overset{\overset{CH_3}{|}}{CH}-CH_2Cl$
 $> CH_3-\overset{\overset{CH_3}{|}}{C}-CH_2Cl$

M.O.F

N.O.F have higher B.P?

(a) $1^\circ R-X$ (b) $2^\circ R-X$ (c) $3^\circ R-X$ (d) ~~all have same~~

Solubility:

They are insoluble in water but soluble in organic solvents.

However they are slightly soluble in H_2O due to polarity
Order: $R-F > R-Cl > R-Br > R-I$
 $\downarrow$
Can't form H-bonding.

Density

Alkyl halides are less dense than H_2O , therefore they float on its surface.
Alkyl iodide & alkyl bromides are denser than H_2O therefore water floats on their surface.

* Density of $R-F$ & $R-Cl$ < Density of H_2O

* Density of $R-I$ & $R-Br$ > Density of H_2O

Surface tension

Size of Halogen $\propto$ S.T

Size of Alkyl gp $\propto$ S.T

Viscosity

Size of Halogen & Alkyl gp $\propto$ Viscosity

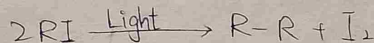
* No. of Branches $\propto$ Viscosity

Specific Gravity

Size of Halogen & Alkyl gp $\propto$ Specific gravity.

They are in general toxic compounds & bring unconsciousness when inhaled in large amounts.

Alkyl halides become violet or brown in colour on exposure to light as they decompose in light.



Alkyl halide burn on copper wire with green flame (Beilstein test for halogen)

They are colourless & odourless.

MCQ

All alkyl halides are insoluble in water except;

(a) $R-F$ (b) $R-Br$ (c) $R-Cl$ (d) $R-I$
Due to hydrogen bonding.

Preparation of alkyl Halide

(1) From Alcohol:

(a) Rxn of Alcohol with HX :

Alcohol react with HX and form alkyl halide in which OH of alcohol is replaced by halogen.

It is an nucleophilic substitution rxn (S_N2 mechanism)

Order of reactivity of HX with alcohol increases down the group due to decrease in bond strength.

* $HI > HBr > HCl > HF$

The HI & HBr react with alcohol without catalyst while HCl react in the presence of anhydrous $ZnCl_2$ catalyst.

HF is not used as it requires more energy to break bcz of its high bond energy.

Order of reactivity of alcohol with HX increase with in degree of alcohol.

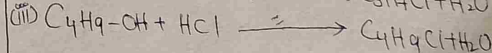
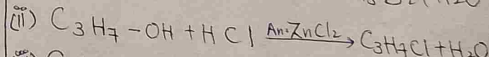
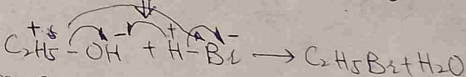
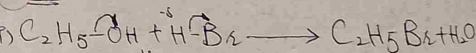
$3^\circ \text{alc} > 2^\circ \text{alc} > 1^\circ \text{alc} > \text{MeOH}$

The solution of $ZnCl_2$ in conc. HCl is called Lucas Reagent.

Used to distinguish.

Only 3° alc react with HCl in the absence of catalyst.

Rxn:



The alkyl halide formed will be according to the degree of alcohol used.

ie $1^\circ-OH \rightarrow 1^\circ-R-X$

$2^\circ-OH \rightarrow 2^\circ-R-X$

$3^\circ-OH \rightarrow 3^\circ-R-X$

$CH_3-OH \rightarrow CH_3-X$

Alkyl halides are the halogenated product of alcohol.

Limitations:

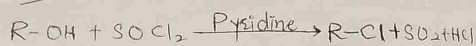
(i) Some side products are formed due to rearrangement of carbocation.

(ii) Formation of alkene due to dehydration of alcohol.

(iii) Rxn of Alcohol with SOCl_2 :

Alcohol react with SOCl_2 to give alkyl halide.

G.Rxn:



Pyridine being a base is used to absorb the HCl produced in the rxn.

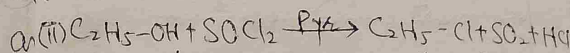
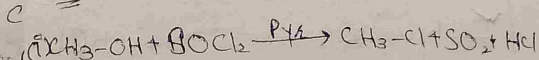
This process is also called ① Darzen's Process

This is ~~an~~ best method to produce alkyl halide bcz the SO_2 & HCl

being gases leave the rxn mixture & left the

product pure.

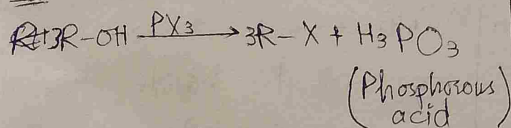
& Ex:



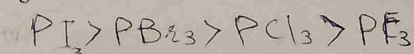
(iii) Rxn of Alcohol with PX_3 :

Alcohol react with PX_3 to give alkyl halide.

G.Rxn:



Reactivity of PX_3 with alcohol increases down the group.



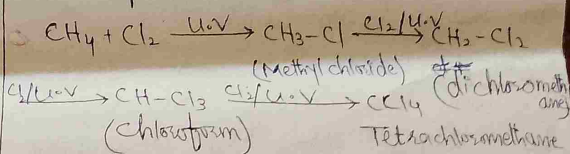
For 1 mole of PX_3 , 3 mole of alcohol are required to produce three mole of

alkyl halide.

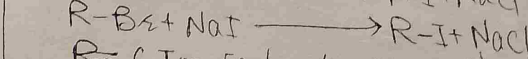
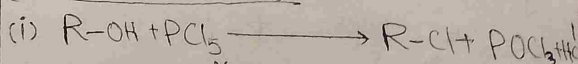
PI_3 & PB_{33} are produced in "situ" (in rxn mixture) by the reaction of red phosphorous with iodine & Br_2 respectively. Some reactants are produced in rxn mixture bcz they are produced outside they may be toxic or unstable.

(R) From Alkane's Halogenation:

Alkane's undergo free radical substitution in the form of halogenation to give alkyl halide. in the presence of UV light. It is not a best method to produce ^{pure} alkyl halides as other haloalkanes are also produced in this rxn.



(3) Other Methods:



R = (Ion-Exchange Method)

W.O.F is the best method of preparation of alkyl halides.

(a) Halogenation of alkane

(b) Rxn of Alc. with SOCl_2

(c) Rxn of Alc. with HX

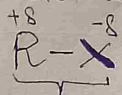
(d) Rxn of Alc. with PX_3

Reactivity of Alkyl Halides:

Alkyl halide are reactive due to polar C-X bond.

The polarity of C-X bond is due to high E.N of halogen atom.

The α -carbon of alkyl halide is electrophilic in nature.



Alkyl halides undergo two types of rxns.

(i) Elimination rxn.

(ii) Nucleophilic substitution rxn.

Halogen is a good leaving group so it takes its electron and removed as stable halide ion.

When a nucleophile replaces halide ion on α -carbon of alkyl halide, it undergoes S_N rxn.

When halide ion along with other atom (often β hydrogen) is removed, alkene is formed & it is termed as elimination rxn.

Elimination & S_N rxn competes with each other & its depends upon attacking ~~nucleophile~~ ^{agent}.

$\Rightarrow$ If attacking agent is Nu^- then S_N rxn will occur.

$\Rightarrow$ If attacking agent is base then elimination rxn will occur.

Characteristic rxns of alkyl halides are S_N rxns.

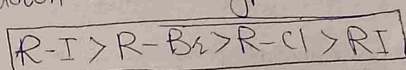
Order of Reactivity

(a) On the Basis of Bond Strength

$\Rightarrow$ Bond strength $\propto$ Reactivity

$\Rightarrow$ Down the gp, bond strength of C-X decreases due to increase in size of halogen atom.

$\Rightarrow$ So reactivity of R-X increases down the gp.



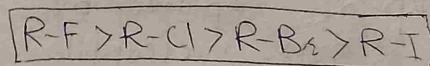
R-X	Bond Energy
C-F	467 kJ/mol
C-Cl	346 kJ/mol
C-Br	290 kJ/mol
C-I	228 kJ/mol

(b) On the Basis of Polarity:

$\Rightarrow \mu \propto \text{polarity} \propto \text{Reactivity}$

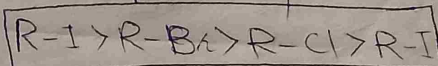
$\Rightarrow$ Down the gp, polarity decreases, so reactivity down the gp decreases.

$\Rightarrow$ Order of Polarity & reactivity is



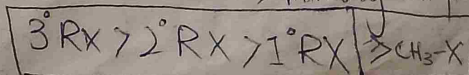
R-X	$\Delta \text{E.N}$	μ
C-F	1.5	1.83 D
C-Cl	0.5	1.87 D
C-Br	0.3	1.81 D
C-I	0.061	1.62 D

But experimentally (main order)



(c) On the Basis of Nature of halogen atom.

Tertiary alkyl halide > Secondary alkyl halide > Primary alkyl halide



(c) Oxidation of Alc. with KMnO_4 and $\text{K}_2\text{Cr}_2\text{O}_7$ in presence of H^+ ions.

more of 3 required

W.O.F is more reactive?

- ☺ Main factor that describes best the reactivity of RX is bond energy.

- "Molecule in which leaving group is replaced by ~~an~~ stronger nucleophile".

- Leaving Group (LG):

OR
The group or specie which is coming replaced by an another nucleophile".

⑩ For SN s_{x_n} , the incoming μ should be stronger than

Leaving power $\propto$ Basicity of gp

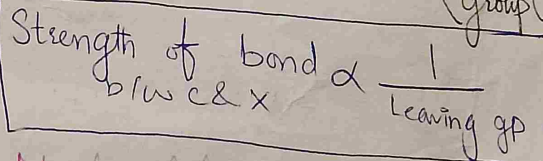
$$I^- > Br^- > Cl^- > F^-$$

→ L. power decreases

* Stability of L. GP $\rightarrow \propto \frac{\text{size}}{EN}$

Excellent	 , NH_3
Very Good	I^- , H_2O
Good	Br^-
Fair	Cl^-
Poor	F^-
Very poor	OH^- , NH_2^- , RO^-

80 E x 00



Nucleophile

ϕ is represented by Nu.

It can
substit
Streng
nucleo
of h
is s

③ Nuc

As we
left
nuclei
e.g.
=

Any high density proton

Rep

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42

Order

★ $\bar{C}H$

★ F

Order

★ 2

A T

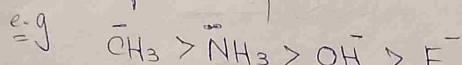
the
d be
ing group.
ity of gp

It undergoes nucleophilic substitution rxn.

Strength of nucleophile is called nucleophilicity & is the measure of how fast the leaving gp is replaced by a nucleophile.

$$\text{Nucleophilicity} \propto \frac{1}{E \cdot N}$$

As we move towards the left of periodic table, nucleophilicity decreases.



Base:

* Nucleophilicity $\propto$ size of molecule/atom
 $\propto$ steric hindrance $\propto$ Resonance
 $\propto$ -ve charge

Any specie that has high affinity for proton & donates its electron pair to proton is called Base.

Represented by $\ddot{\text{B}}$.

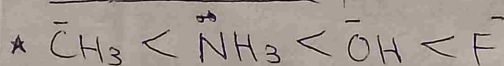
It undergoes elimination rxn.

Strength of base is called basicity & is measured in terms of equilibrium constant for abstraction of proton.

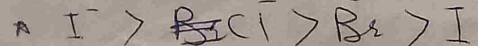
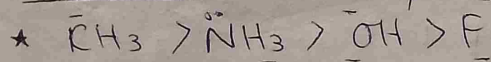
$$\text{Basicity} \propto E \cdot N$$

$$\text{Basicity} \propto \frac{1}{\text{Nucleophilicity}}$$

Orders For Basicity:



Orders For Acidity/Nucleophilicity

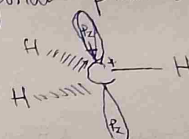


Carbocation is an ion in which carbon bears positive charge & three bonds is called carbocation.

Important Pts:

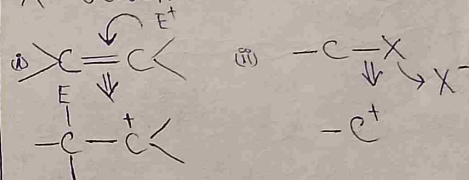
Carbon in carbocation is sp^2 hybridized.

Trigonal planar structure.



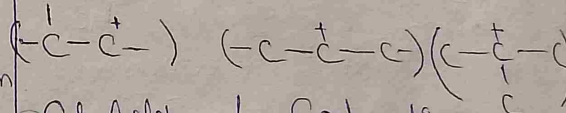
Diamagnetic.

These are formed when an E^+ attack on alkene or X detaches as halide ion.



Types of Carbocation:

1° Carbocation 2° Carbocation 3° Carbocation



Stability of Carbocation:

Stability of carbocation depends upon the stabilization of +ve charge of carbon of carbocation.

If there are more ED groups there will be more stabilization of +ve charge of carbon & carbocation will be more stable.

When alkyl gps are attached to carbocation's carbon, they will stabilize the carbocation's

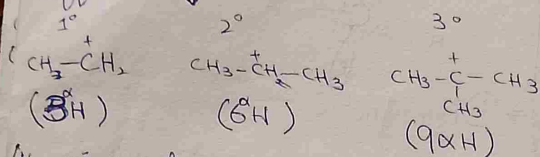
Carbon +ve charge & ^{will} make the carbocation stable.

No. of alkyl gps $\propto$ Stability of carbocation

Order of Stability:

Methyl carbocation cation $<$ 1° carbo- $<$ 2° $<$ 3° carbocation carbocation

Alkyl gps donate electrons to carbon atom through hyperconjugation & inductive effect.



(Hyperconjugative α -hydrogen $\propto$ stability)

Nucleophilic Substitution Rxns

Those rxns in which one nucleophile is replaced by another nucleophile is called SN rxn.

Alkyl halide undergo two different mechanisms. The two different mechanism depends upon the nature of alkyl halide.

SN₁ Rxn:

It stands for unimolecular nucleophilic substitution.

The type of SN rxn in which C-X bond breaking & C-Nu bond formation occurs one after other in two different steps is called unimolecular nucleophilic substitution (SN₁ rxn).

Ex^o Characteristics:

It is a two step process.

Generally tertiary alkyl halide follow this mechanism.

This occurs in the presence of polar protic solvents such as CH_3-OH , C_2H_5-OH , CH_3COOH , H_2O etc.

This mechanism is followed when weak nucleophile is used.

First step of this rxn is slow & the second step is fast.

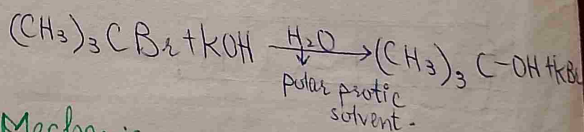
The molecularity of this rxn is one bcz only one molecule is involved in its RDS.

Two transition states are formed in this mechanism.

During formation of intermediate

During intermediate $\rightarrow$ product

Ex:

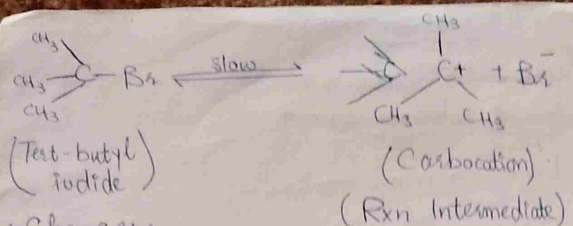


Mechanism:

Step No 1: Ionization:

The first step of this rxn is ionization of alkyl halide to form carbocation.

It is done in the presence of polar protic solvent bcz of its high ionizing power.

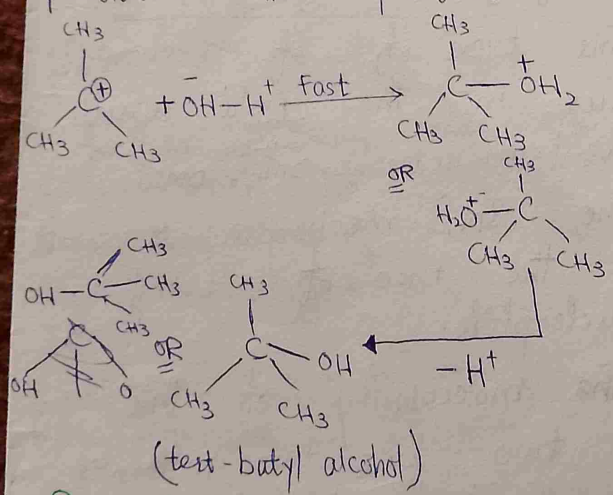


→ Changes:
 * $sp^3 \rightarrow sp^2$
 * Tetrahedral → Trigonal planar
 * $109.5^\circ \rightarrow 120^\circ$

* High steric hindrance → Low steric hindrance
 * Weak E^+ character → Strong E^+ character
 ⇒ An nucleophile can't attack alkyl halide without ionization due to:
 * Large steric hindrance
 * Weak E^+ character of carbon (Due to sp^3 hybridized)

Step 2: Attack of Nu:

A Nu attack on E^+ carbon & form a transition state which upon release of H^+ produces the product.



Rxn Kinetics:

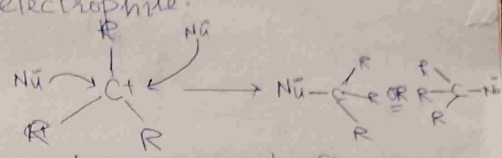
(14) The kinetics of this rxn shows that $SN1$ mechanism follows 1st order rxn.

$$\text{Rate} = k[(CH_3)_3CBr]$$

⇒ Rate of rxn depends upon conc. of alkyl halide.

Direction of Attack:

(15) In $SN1$ rxn mechanism, an nucleophile can attack on both sides of carbocation.
 (16) There are equal chances for two sides attack of electrophile.



Configuration of Product:

(17) In $SN1$ mechanism, there is 50% retention & 50% inversion.
 (18) When a nucleophile attack on E^+ on the side of leaving it is called retention of configuration.
 (19) When a nucleophile attack on E^+ on the opposite of leaving gp it is called inversion of configuration.

Racemization

(20) If a reactant is chiral, mixture of two enantiomers is produced called racemic mixture & this process is called racemization.

(21) Racemization is the combination of inversion & retention.

(22) In $SN1$ mechanism, rarely the product is completely racemic mixture rather there is more inversion than retention. This is called partial racemization.

Rate:

Rate of $SN1$ rxn doesn't depend on:
 (a) Conc. of $R-X$ (b) Conc. of Nucleophile
 (c) Nature of leaving gp (d) Stability of carbocation
 (e) Nature of solvent.

Imp. Pts:

→ SN1 is exothermic in nature. ($\Delta H < 0$)

→ SN1 rxn is done at low Temp

→ Factors on which SN1 rate depends are

(i) Nature of Alkyl halide (R-O-X) (R-O-X is R-O-X)

(ii) Nature of leaving group

→ easily leaving $\uparrow$
($I > Br > Cl > F$)

(iii) Nature of solvent

(Solvent Dielectric $\propto$ Rate of Rxn)

e.g. $H_2O > C_2H_5OH > CH_3COOH$
Dielectric (80) (35) (5)

MCO:

Correct order of reactivity of Rxn in SN1 mechanism is -

(a) $1^\circ > 2^\circ > 3^\circ$ (b) $3^\circ < 2^\circ < 1^\circ$

(c) $1 < 2 < 3$ (d) $3^\circ > 2^\circ > 1^\circ$

MCO:

The unit of rate constt 'k' in SN1 mechanism is -

(a) $\text{mol dm}^{-3} \text{s}^{-1}$ (b) $\text{mol}^2 \text{dm}^{-6} \text{s}^{-1}$
(c) s^{-1} (d) $\text{mol}^2 \text{dm}^{-6} \text{s}^{-1}$

MCO:

In case of SN1 mechanism, if conc of both nu & tert. alkyl halide is $\uparrow$ by 4 times then ROR becomes

(a) 8X (b) 16 times
(c) 4X (d) Remain the same

MCO:

Hybridization

Shapes changes during SN1 mechanism are:

(a) $sp^3-sp^3-sp^3$ (b) $sp^3-sp^2-sp^2$
(c) $sp^2-sp^3-sp^2$ (d) No changes

MCO:

→ Shapes changes:

Tetrahedral $\rightarrow$ Trigonal planar $\rightarrow$ Tetrah.

→ Angle changes:

$109.5^\circ \rightarrow 120^\circ - 109.5^\circ$

→ Steric hindrance changes:

more $\rightarrow$ Less $\rightarrow$ more

→ E⁺ of carbon changes:

Less $\rightarrow$ more $\rightarrow$ Less

SNR Mechanism:

The type of SN rxn in which C-X bond breaking & C-Nu bond formation occur at the same time & same step is called SNR mechanism.

Character:

It is generally followed by 1° R-X & methyl R-X

It is a single step process

No rxn intermediate (carbocation) is formed during this rxn.

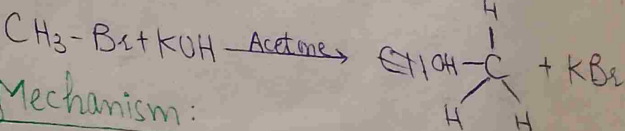
Only one transition state is formed during this rxn.

This type of mechanism occurs in the polar aprotic solvent such as acetone, DMF, DMSO, $CH_3-C(=O)-CH_3$, $CH_3-C(=O)-CH_2-CH_3$

The SNR mechanism occurs in the presence of strong nucleophile.

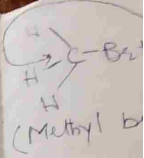
The molecularity of this rxn is two bcz two molecules are involved in RDS. No rearranging occur.

EX:



Mechanism:

→ This is a single step process which occurs as follows



→ The extent

C-X. Rxn

rate both these rxn.

→ R

① is

from size by Nu

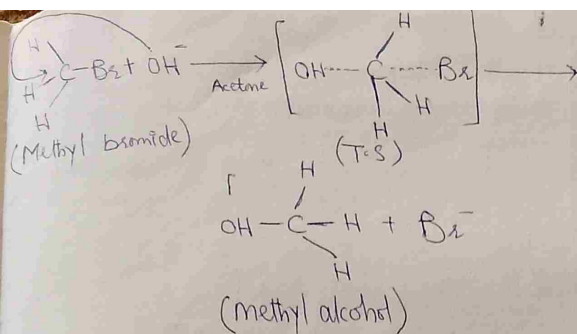
Cont

③ In 100

Ei.e oppo

Imp.

→ In in carbo



⇒ The extent of bond formation b/w C-Nu is equal to the extent of bond breaking b/w C-X.

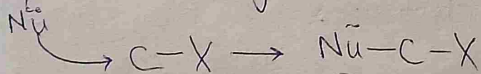
Rxn Kinetics:

According to rxn kinetics, the rate of rxn depends on both the conc. of nu & substrate.
 ⇒ SNR follows 2nd order rxn.

$$\text{Rate} = k[\text{CH}_3\text{-X}][\text{Nu}^-]$$

Direction of attack:

A nu can only attack from back side bcz front side is occupied & hindered by halogen.



Configuration of Product:

In SNR mechanism, there is 100% inversion of product (i.e. product attack from the opposite side of halogen).

Imp. Pts:

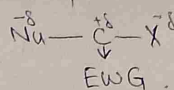
In SNR mechanism, Nu donate e⁻ in the antibonding orbital of carbon atom.

⇒ SNR mechanism is favored at low temp.

⇒ Factors on which SNR

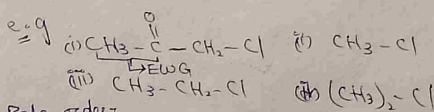
rate depends are;

- (i) Nature of Substrate (1° > 2° > 3°) (Rate ∝ steric hindrance)
- (ii) Nature of Nu.
- (iii) Stability of Transition state.

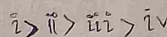


(Stability of T.S. ∝ Rate)

*T-state factor dominates steric hindrance.



Rate order:



in Nature of solvent.

polar aprotic solvent.

Meq:

Tertiary alkyl halide are inert towards SNR mechanism due to:
 (a) Inductive effect (b) Insolubility
 (c) Steric hindrance (d) All of these

ELIMINATION RXN

Alkyl halide undergoes a type of elimination rxn called β-elimination rxn.
 1,2-elimination rxn.

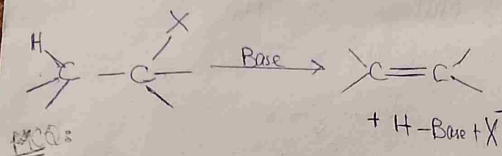
β-elimination Rxn:

The reaction in which two atoms are removed from two adjacent C-atoms (α & β) is called β-elimination rxn.

Alkyl halide upon β-elimination in the presence of strong base give alkene.

For β -elimination, 4nce of β -hydrogen is necessary.

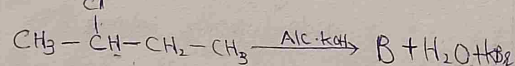
β -Hydrogen in alkyl halide is slightly acidic due to EW effect halogen.



$\text{CH}_3\text{-Cl} + \text{Base} \xrightarrow{\text{Alc.}} \text{'A'}$. The product 'A' is

- (a) Alkane (b) Alkene
(c) Alcohol (d) No rxn
(There is no β -H)

The product of the rxn is



- (a) $\text{CH}_2 = \text{CH} - \text{CH}_2 - \text{CH}_3$ (b) $\text{CH}_2 = \text{CH} = \text{CH} - \text{CH}_3$
(c) $\text{CH}_3 - \text{CH} = \text{CH} - \text{CH}_2 - \text{CH}_3$ (d) No rxn

Saytzeff's Rule

When there is are two β -H's. then the base will attack on the carbon atom that have less no of H-atoms.

E1 Mechanism:

Two steps elimination rxn in which leaving gp leaves first followed by removal of hydrogen atom is called E1 rxn.

Character:

- (1) It is generally followed by tertiary alkyl halide.
(2) It is just like a two step process.
(3) It occurs in the rxn of weak base.

It occurs in the rxn of polar protic solvents.

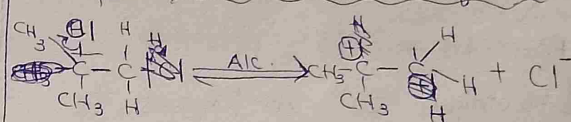
The molecularity of this rxn is one bcz only one molecule is involved in its RDS.

Rxn intermediate formed during this rxn is carbocation.

There are two transition states in E1 mechanism. Mechanism: It competes with SN_1 rxn. Rearrangement can occur.

E1 completes in two steps.

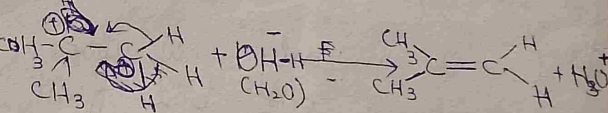
Step 1: Ionization/Formation of Carbocation



- * Slow step
- * RDS
- * Reversible
- * Heterolytic cleavage of C-X bond occurs.
- * Similar to that of SN_1 .

Step 2: Attack of Base

Base attack on β -hydrogen & abstract it from the substrate to form double bond.



- * Fast
- * Irreversible
- * Formation of product occurs.

Rxn Kinetics:

The molecules in RDS is only one so the order of rxn is one i.e it is 1st order rxn.

$$\text{Rate} = k[(\text{CH}_3)_3\text{C-Cl}]$$

Rate of rxn only depends on conc. of alkyl halide.

Rate of rxn doesn't depends on conc. & nature of base.

- W.O.F has same 1st step?
(a) SN_2 & E2 (b) SN_1 & E2
(c) SN_1 & E1 (d) E2 & E1

MCQ:

Rxn in which rearrangement can occur & result in polar product is —

(a) E₁ (b) E₂ (c) SN₁ (d) SN₂

MCQ:

E₂ mechanism

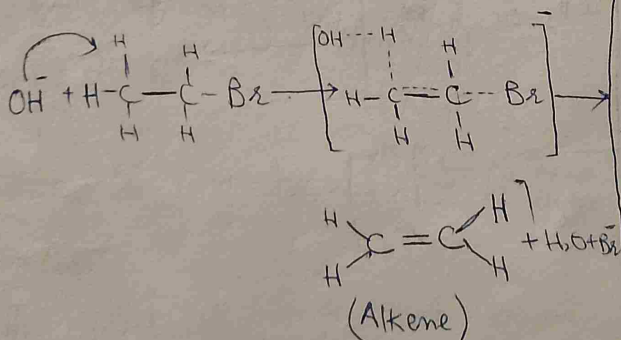
(5) The elimination rxn in which formation of double bond & removal of leaving group occur at the same time. is called E₂ mechanism.

Properties

- (5) It is two step process.
- (5) Generally followed by primary R-X.
- (5) Occur in polar aprotic solvent.
- (5) No transition rxn intermediate is formed. Only one transition state exists.
- (5) Occur in the +ve of strong base.
- (5) Molecularity of rxn is two.
- (5) No rearrangement can occur.
- (5) Competes with SN₂ rxn.

Mechanism:

(5) It occurs in single step.



Rxn Kinetics:

(5) It follows 2nd order rxns

$$R = k[R-X][Base]$$

MCQ: B-elimination can occur when
 (a) Nu attacks α -hydrogen (b) Nu attack B-hyd-
 (c) Nu attacks α -carbon (d) Nu attacks B-carbon

Imp. Pts about Elimination Rxn

- Endothermic ($\Delta H > 0$)
- Spontaneous/feasible at high Temp.
- Reverse of addition rxn
- Saturated (Alkyl halide) $\rightarrow$ Unsaturated (Alkene)
- ⇒ Rate of Elimination Rxn $\propto$ Ability of LG (Ab. of LG $\uparrow$, Rate of E Rxn $\uparrow$)

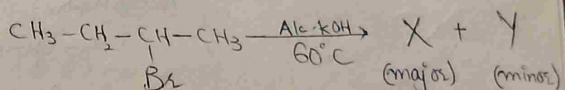
MCQ:

R^o R-X can show — rxns

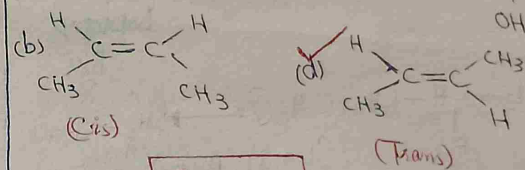
(a) SN₁ (b) SN₂ (c) E₁ (d) E₂ (e) All of these

MCQ:

Identify the major product (X)



(a) CH₃-CH₂-CH=CH₂ (b) CH₃-CH₂-CH(OH)-CH₃



Trans > Cis

Substitution Vs Elimination

(5) R-X can undergoes both Nu sub & elimination rxn. Both these rxns compete with each other.

Factors Deciding SN & E rxn:

(5) There are certain factors that decide whether an Elimination or nucleophilic rxn will occur.

(i) Nature of alkyl halide:

- ⇒ more crowded alkyl halide favours E rxn over SN rxn.
- ⇒ It is bcz of steric hindrance that protect the α -carbon from Nu attack.

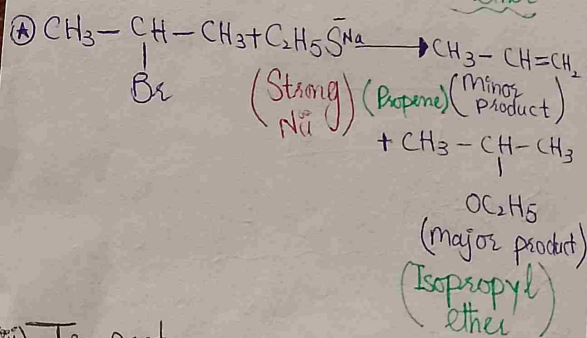
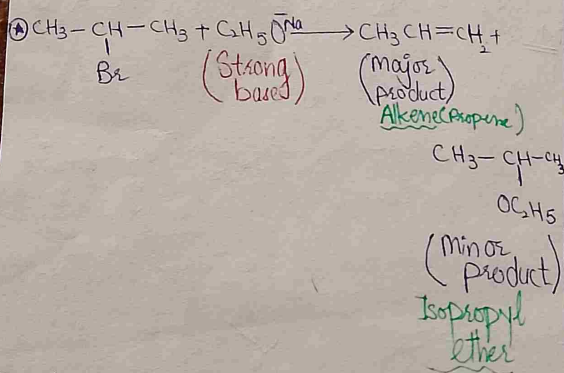
⇒ Another reason is that the more substituted product R-X stabilizes the alkene more than substitution product.

(iii) Strength of Base or Nu:

⇒ If the attacking reagent is a strong base then elimination rxn will occur.

⇒ If the attacking reagent is a strong nucleophile then nucleophilic substitution will occur.

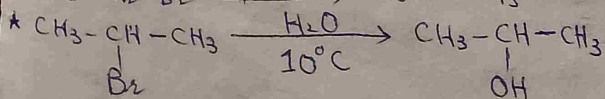
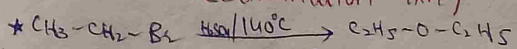
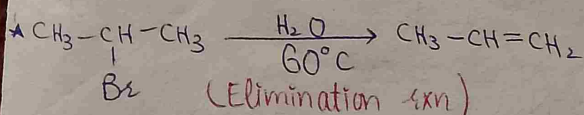
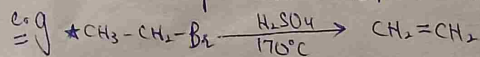
e.g



(ii) Temperature:

⇒ High temperature → Elimination rxn

⇒ Low temperature → SN rxn



(N. Substitution rxn)

(iv) Site of Attack:

⇒ If a Nu⁻ attack on E⁺ carbon then SN rxn occur.

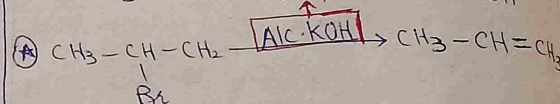
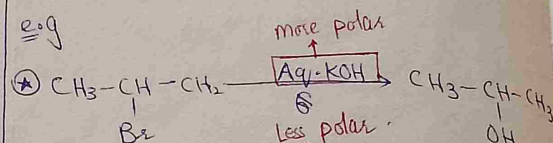
⇒ If a Nu⁻ attack on E⁺ hydrogen or β-hydrogen then elimination rxn will occur.

(v) Nature of Solvent:

⇒ If solvent is more polar then SN rxn will occur.

⇒ If solvent is less polar the elimination rxn will occur.

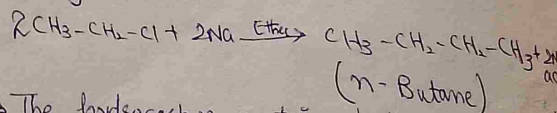
e.g



Other Rxns of R-X

(i) Wurtz Synthesis:

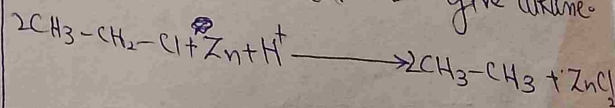
⇒ R-X reacts with Na in dry ether to give symmetrical alkanes.



⇒ The hydrocarbon contains double the no. of C-atoms that in R-X.

(ii) Reduction of Alkyl halide:

⇒ R-X on reduction with Zn in the presence of an aqueous acid such as HCl or CH₃COOH to give alkane.



Organometal

⑤ All the which contain one C-metal structure compounds.

⑥ They are by Ar-CuLi

⑦ Their exo

* Silver ac

* Tetrameth

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(a)

Organometallic Compound

All the organic compounds which contain at least one C-metal bond in their structure are called organometallic compounds.

They are generally represented by $Ar-M$ or $R-M$.

Their examples are;

- * Silver acetylides
- * Grignard reagent
- * Tetramethyl lead
- * Dimethyl zinc etc.

They are named simply by writing the name of alkyl/aryl gp followed by the name of metal.

Grignard Reagent

The G.R. was discovered, study & prepared by Victor Grignard, for this we also won Nobel prize in 1912.

Grignard reagent is represented by $R-MgX$.

The R may be

- Aryl
- Alkyl
- Vinyl
- Allyl

The X may be

- Br
- Cl
- I

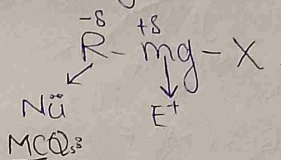
(No F more reactive)

Meo:

The part of G.R. that remain constt in all its compounds is

- (a) R (b) mg (c) X (d) All

They are highly reactive compounds and are so important bcz of its large synthetic application due to polar $R-Mg$ bond.



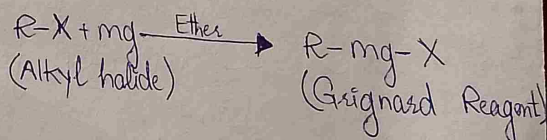
$R-Mg$ = Covalent bond

$Mg-X$ = Ionic bond.

Preparation of Grignard Reagent

Grignard reagent are prepared by rxn of alkyl halide with freshly prepared magnesium in dry ether. i.e. Diethyl ether or tetrahydrofuran.

Rxn:



It is an example of heterogeneous rxn bcz the $R-X$ react with solid Mg in ether.

$R-MgX$ can't be isolated. It is used as ethereal solution in rxn condition.

Ease of Formation of G.R.:

Ease of formation of G.R.

depends on;

- (i) nature of R or Ar gp
- (ii) nature of halogen atom.
- (iii) Nature of Alkyl/Aryl gp:

With increase in length of alkyl or aryl group the

ease of formation of G.R decreases.

Length of Alkyl & Stability of G.R
 ↳ Aryl gp

(ii) Nature of Halogen Atoms

For same alkyl group, the ease of formation of G.R ↑ increases down the gp.



The most commonly used is alkyl magnesium bromide bcz it is less toxic & less expensive than alkyl magnesium iodide.

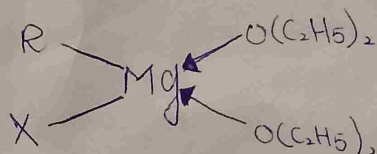
Role of ether:

(i) Act as a medium.

(ii) Stabilize the G.R by solvation.

The dietherate solution of G.R is crystalline in which Mg is co-ordinated with two ethers & an alkyl gp & an halogen atom.

The dietherate sol. of G.R exist in tetrahedral geometry.



Reactivity of Grignard Reagent

C-Mg Bond in G.R is highly polar due to greater E.N difference b/w C-Mg. Due to this partial -ive & partial +ve charge comes on carbon & Mg respectively. This makes the carbon an e⁻-rich species & can attack on electron deficient centers

i.e. can act as Nu⁻.

Characteristic rxns of G.R are nucleophilic substitution rxn & nucleophilic addition rxns.

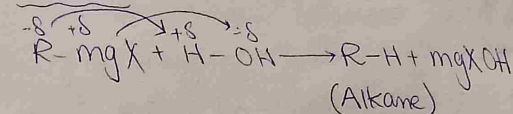
Nu Substitution Rxn of G.R

(i) Rxn With Water:



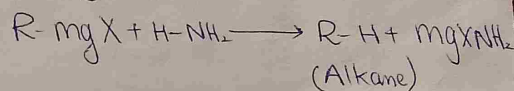
Grignard reagent on rxn with H₂O gives alkane.

Mechanism:



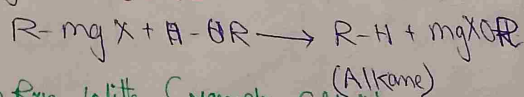
(ii) Rxn With Ammonia:

Grignard reagent on rxn with NH₃ gives alkane.



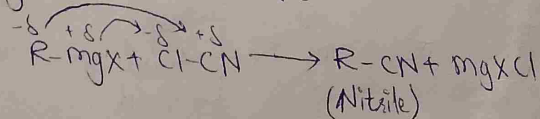
(iii) Rxn With Alcohol:

G.R Alcohol upon rxn with alcohol gives alkane.



(iv) Rxn With Cyanogen Chloride:

G.R Alcohol upon rxn with Cl-CN gives nitriles.



As Rxn With Aldehyde & Ketone:

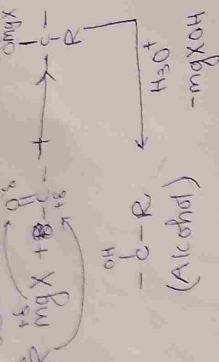
As G.R reagent react with aldehyde & ketone and gives an addition product which upon acid workup gives alcohol.

Formaldehyde $\rightarrow$ 1° alcohol

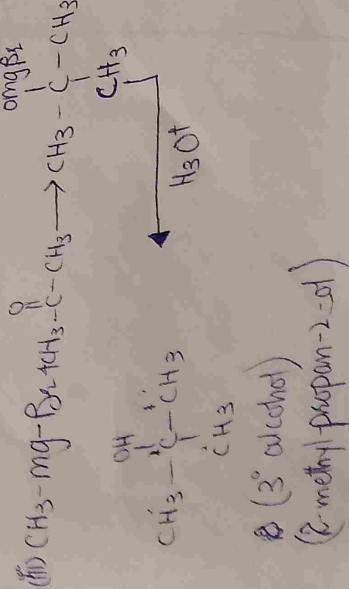
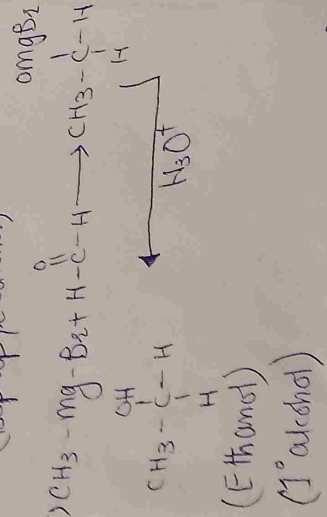
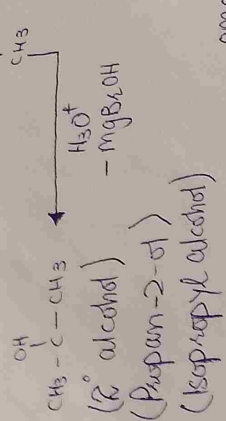
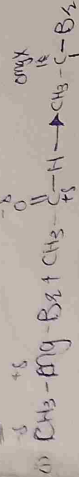
Higher aldehyde $\rightarrow$ 2°

Ketone $\rightarrow$ 3°

Mechanism:



Ex:



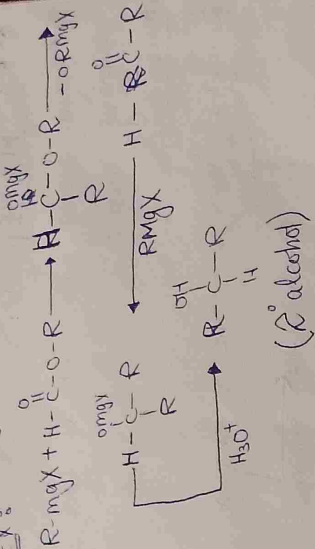
As Rxn With Esters:

As G.R on rxn with an addition product which upon elimination gives aldehyde or ketone. Aldehyde or ketone on rxn with another G.R yield alcohol. As rxn with formate ester:

As G.R on rxn with formate ester yield alcohol at the end.

This process produces only secondary alcohol.

Mechanism:

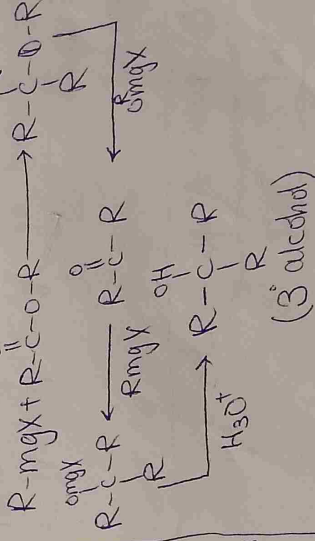


As Rxn With Ester of higher C-Acid:

G.R reagent on rxn with ester of higher carboxylic yields alcohol.

This rxn only produces tertiary alcohols.

Mechanism:



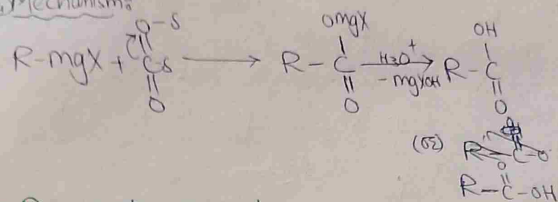
As Rxn With CO₂:

G.R upon rxn with CO₂ gives carboxylic acids.

⑤ This method is particularly used for the conversion of RX to carboxylic acid.

⑥ The carboxylic acid formed contains one carbon atom more than the corresponding primary alkyl halide. So this rxn can be used to extend the length of product.

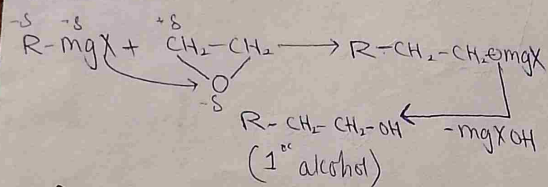
⑦ Mechanism:



⑧ Rxn With Epoxide:

⑨ Epoxide reacts with G.R to give primary alcohol.

⑩ Mechanism:



MCO:

W.O.F on rxn with G.R gives straight chain alcohols?

(a) Epoxide (b) Formaldehyde

(c) Acetaldehyde (d) Both (a) & (b)

MCO:

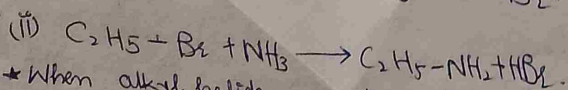
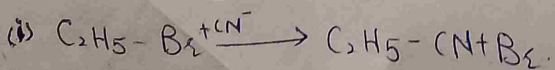
W.O.F on rxn with G.R doesn't give alcohol as final product?

(a) Esters (b) CO_2

(c) Ketone (d) Both (a) & (b)

Extra
MCO From Alkyl halides

From $SN1$ rxn Topic:



* When alkyl halide reacts with excess of NH_3 then Quaternary alkyl ammonium ion will be formed.

* When NH_3 reacts with excess of alkyl halide then primary amine will form.